Functional Test Plan — development build 0.1.8 (build 9)

> Builds 4 and later add scanning; scan tests are in `docs/task4c/PIXEL_SCAN_TEST_GUIDE.md`, and the result-screen checks R1–R10 in its section 4a. The typed-entry tests below apply through *Type the details instead*. No Android phone? Use `docs/TEST_ON_COMPUTER.md`.

Purpose: check, on a real Android phone, that the app does what it should, and that its wording never goes beyond the evidence. Install first: `docs/INSTALL_ON_ANDROID.md`.

Status on a physical device: NOT RUN. No phone or emulator is available in the build environment. The "Automated" column shows what has been exercised automatically on the real screens (`mobile/__tests__/flows.test.tsx`, recorded real FDA/NLM responses) or on the shared logic (`npm test`). Automated tests do not replace the device test.

Test data (public, non-personal)

Use only these. They are public product codes from FDA's NDC Directory; results come from FDA's live data, so a later FDA update can change an expected result — note it and continue.

ID	Medication (type exactly)	NDC	Strength	Form	Lot	Why
D1	METFORMIN HCL ER	62756-142-02	500 MG	TABLET	JKU4639A	Product with a known recall (D-0325-2021) whose lot list contains JKU4639A
D2	METFORMIN HCL ER	62756-142-02	500 MG	TABLET	*(blank)*	Same product, no lot → possible match
D3	METFORMIN HCL ER	62756-142-02	500 MG	TABLET	ZZZ0000	Lot not in the recall list
D4	ROSUVASTATIN CALCIUM	82009-018-10	10 MG	TABLET		No applicable recall in the data recorded 2026-11-12
D5	WARFARIN SODIUM	67046-2072-3	1 MG	TABLET		Excluded by policy (blood thinner)
D6	METOPROLOL TARTRATE	*(blank)*	25 MG	TABLET		Not covered by this version
D7	METFORMIN	00000-0000-00	500 MG	TABLET		NDC that does not exist
D8	FUROSEMIDE	68094-756-62	10 MG/ML	ORAL SOLUTION		Repackaged product (source may not be established)

Never type your own medication, your name or a prescription number.

How to tell the four results apart

Every result has a label at the top of the result box, a headline, and next steps:

Label	Headline starts with	Means	Never means
Recall match found	"Available recall information matches the medication details you provided."	A recall record names this product and its conditions (e.g. your lot) appear to apply	That FDA verified your supply
Possible recall match	"This medication may be affected by a reported recall…"	A recall might apply; something (often the lot) is unknown	That it does not apply
No matching recall found in the FDA records checked	"No matching recall was identified in the sources successfully checked as of [date]. This does not establish…"	Every search completed and nothing matched *in those records*	That the medication is safe or recall-free
Could not determine	"We could not reliably determine whether this medication is affected by a recall."	Something failed, was incomplete, too old or inconsistent	A no-match

No result is ever shown in green or with a check mark.

Tests

Result column: write PASS / FAIL / NOTE for the device run.

#	Area	Steps	Expected	Automated	Device
F-01	Install & launch	Install build 9, open	"MedLuma (DEV)"; purple DEVELOPMENT BUILD / evaluation prototype banner; "Before you start"	UI: first launch	
F-02	Legal documents	On "Before you start", open Terms, Privacy, Medical Disclaimer; Back	Each opens, marked draft; Back returns	UI: first launch	
F-03	Version	About and privacy	"version 0.1.8 (build 9)"	Unit: version	
F-04	Manual entry & identification	Enter D1, *Find this medication*	Spinner "Looking up your medication…", then *Is this your medication?*	UI: confirmed recall	

#	Area	Steps	Expected	Automated	Device
F-05	Evidence labels	On the confirmation screen	Product, ingredient, form, release, labeler: *From FDA records*; NDC note "A listing is not FDA approval..."; labeler note; lot: *Entered by you...* with "Turn on the lot number item below..."	UI: confirmed recall	
F-06	Confirmation required	Press *Yes, these details match my label* before switching every item on	Button stays disabled until every required item is on	UI: confirmed recall	
F-07	Confirmed recall incl. lot	Switch all items on (including the lot), confirm	Spinner "Searching FDA recall records..."; label **Recall match found**; D-0325-2021 under *Recalls that match your product*; "Your lot number appears in this recall's list of lots."; next steps incl. pharmacist and "Do not stop, change or replace..."	UI: confirmed recall	
F-08	Source attribution & time	On the F-07 result	*What we checked*: "Retrieved on this phone: [time]", "FDA recall data last updated: [date]", searches completed, search check passed, "not updated automatically", "this medication only"	UI: confirmed recall	
F-09	Official record	Tap *Official FDA record* under D-0325-2021	Browser opens api.fda.gov showing that recall	UI: confirmed recall	
F-10	Possible match (no lot)	D2 (do not enter a lot)	**Possible recall match**; "You did not enter a lot number" qualifier	UI: possible recall	
F-11	Lot not listed	D3, switch on the lot item	Not *Recall match found*; record text "Your lot number was not found in the listed lots. That does not mean your supply is unaffected."	Engine tests	
F-12	No match	D4	**No matching recall found in the FDA records checked** with date and "This does not establish..."; *Records for a different product* counted, listed under *Show these ... records*; no "safe" anywhere	UI: no match found	
F-13	Excluded	D5	"We don't check this medication... No recall check was done."; rule EXCL-WARFARIN; "Not checking some medications does not mean other medications are safe."	UI: excluded	
F-14	Unsupported	D6	"This medication or form is not covered by this version of the app. No recall check was done."	UI: unsupported	
F-15	Unknown NDC	D7	Box **NDC not found**: "We could not find this NDC in FDA's directory..."; fields *Entered by you, not checked against FDA records*; release question shown	UI: unknown NDC	

#	Area	Steps	Expected	Automated	Device
F-16	Missing fields	Only a name, *Find this medication*	"Please enter the drug name, strength and form."; typed name still there	UI: missing fields	
F-17	Network failure	D4 up to the confirmation screen; turn on airplane mode; confirm	Spinner while it retries (up to ~10 s), then "We couldn't reach FDA's data. Nothing was checked." with *Try again*; never a no-match	UI: network failure	
F-18	Recovery	Turn airplane mode off; *Try again*	A normal result	UI: network failure	
F-19	Repackaged product	D8	Either a result with the qualifier about the original product not being confirmed, or a result naming the verified source; never an unqualified no-match if the source is unknown	Engine/e2e tests	
F-20	Back button	From any result press Android Back	Returns to the entry screen with typed values (does not close the app)	UI: Android Back	
F-21	Change details	On confirmation, *Change details*	Entry screen with typed values kept	UI: change details	
F-22	Encryption self-test	About and privacy → *Run encryption self-test*	Four lines, each PASS: SQLCipher compiled in (shows a version), read back with the right key, unreadable with a different key, test database deleted. **Copy the exact text into the results**	Logic: unit (fakes); UI: display	
F-23	Delete all data	About → *Delete all data on this phone* → *Delete*	Notice "Before you start" shows again; after *I understand*, entry fields are empty	UI: delete all	
F-24	History off	*Saved checks*	"Saving is turned off in this build because storage encryption has not been verified on a device."	UI: history	
F-25	No personal data leaves	(optional, needs adb) `adb logcat -b crash -d` after the tests	Empty or crash lines only; no medication text	—	
F-26	Large text	Settings → Display → Font size: largest; repeat F-04…F-07	No text cut off; all buttons reachable by scrolling	—	

#	Area	Steps	Expected	Automated	Device
F-27	TalkBack	Turn on TalkBack; repeat F-04…F-07	Each field read with its name and source; result label read as a heading; progress announced	—	
F-28	Dark mode	Turn on dark theme	Text readable (the app uses a light theme)	—	
F-29	Rotation	Rotate during F-07	No crash; result still shown	—	
F-30	Uninstall	Uninstall the app	App and data removed	—	

Recording results

Copy this table into `docs/ANDROID_DEVICE_TEST_PLAN.md` → Results, with: phone model, Android version, APK file name and SHA-256, date, and PASS/FAIL/NOTE per test. Report failures as described in `docs/INSTALL_ON_ANDROID.md` ("Report it without personal data").